LECTURE NOTES

Discrete Mathematics notes

Counting

DM549 - Discrete Mathematics

Simon Holm
Fall 2024

SDU
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

Contents

1 Counting Rules

The following rules are nice to remember for counting.

Definition: The Division Rule

Suppose A is a finite set with $A = B_1 \cup B_2 \cup \ldots \cup B_n$ where

- $|B_i| = d \quad \forall i$ and
- $B_i \cap B_j = \emptyset \quad \forall i, j \quad$ where $i \neq j$

Then $n = |A|/d$.

Definition: The Product Rule

For any finite sets $S_1, S_2, \ldots, S_n$

$$\underbrace{\left|\bigtimes_{i=1}^{n} S_i\right|}_{|S_i \times S_2 \times \ldots \times S_n|} = \underbrace{\prod_{n}^{i=1} |S_i|}_{|S_1| \cdot |S_2| \cdot \ldots \cdot |S_n|}.$$

Definition: The Sum Rule (for two sets)

For any finite sets S_1, S_2 with $S_1 \cap S_2 = \emptyset$, it holds that $|S_1 \cup S_2| = |S_1| + |S_2|$.

Definition: The Sum Rule

For any finite sets S_1, S_2 with $S_1 \cap S_2 = \emptyset$, it holds that

$$\underbrace{\left|\bigcup_{i=1}^{n} S_i\right|}_{|S_i \cup S_2 \cup \ldots \cup S_n|} = \underbrace{\sum_{n}^{i=1} |S_i|}_{|S_1| + |S_2| + \ldots + |S_n|}.$$

Definition: The Subtraction Rule

For any finite sets $S_1, S_2, \ldots, S_n$ it holds that $|S_1| \cup |S_2| = |S_1| + |S_2| - |S_1 \cap S_2|$.

2 Pigeonhole

The Pigeonhole Principle (Theorem 6.2.1 from book)

Let $k \geq 1$ be an integer. When $k + 1$ or more objects are placed into k boxes, there exists at least one boz that contains at least two of the objects.

- Intuition: If there exists 10 holes, but 11 pigeons, at least one hole must contain 2 pigeons.

3 Permutations ans combinations

Permutations count ordered selections of r elements from a set S of n elements (order matters).

Permutations 1

If $n > 0 \in \mathbb{Z}$ and $r \in \mathbb{Z}$ such that $1 \leq r \leq n$, then there are

$$P(n, r) = n(n-1)(n-2)...(n-r+1)$$

r-permutations of a set with n distinct elements.

Permutations 2

If $n, r \in \mathbb{Z}$ such that $0 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Combinations

The number of r-combinations of a set with n elements, where $n > 0 \in \mathbb{Z}$ and $r \in \mathbb{Z}$ such that $0 \leq r \leq n$, equals

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Note that $C(n, r)$ can also be written as $\begin{bmatrix} n \\ r \end{bmatrix}$ (r choose r), known as the binomial coefficient 😉.

4 Binomial Coefficient

The Binomial Theorem

Let x and y be variables, and let $n > 0 \in \mathbb{Z}$

Then.

$$(x+y)^n = \sum_{n}^{j=0} \begin{bmatrix} n \\ j \end{bmatrix} x^{n-j} y^j = [n0]x^n + [n1]x^{n-1}y + \ldots + [nn-1]xy^{n-1} + [nn]y^n.$$

Where $[nj] = C(n,j) = \frac{n!}{j!(n-j)!}$

5 Repetition and indistinguishably

Theorem: Repeating combinations

There are $C(n+r-1,r) = C(n+r-1,n-1)$ r-combinations from a set with n elements when repetition is a allowed

	k-distinguishable boxes	k-indistinguishable boxes
n-distinguishable objects	$\frac{n!}{n_1! \cdot n_2! \cdot \ldots \cdot n_k!}$	$\sum_k^{j=1} \frac{1}{j!} \sum_{j-1}^{i=0} (-1)\hat{\imath}[ji](j-i)^n$
n-indistinguishable objects	$C(n+1-1,k)$	No formula

Table 1: Good to remember 😏

	number of r-permutations of S	number of r-combinations of S
without repetition	$P(n, r)$	$C(n, r)$
with repetition	n^r	$C(n + r - 1, r)$

Table 2: Good to remember 2 😏